University of Debrecen, Faculty of Informatics

Logic in computer science

Zero-order logic
(propositional logic)
semantics rules, theorems

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Basic propoerties

Commutative connective: $\circ \in \{\wedge, \vee, \equiv\}$

$$(A \circ B) \Leftrightarrow (B \circ A)$$

Associative connective: $\circ \in \{\wedge, \vee, \equiv\}$

$$(A \circ (B \circ C)) \Leftrightarrow ((A \circ B) \circ C)$$

Idempotent connective: $\circ \in \{\wedge, \vee\}$

$$(A \circ A) \Leftrightarrow A$$

Negation

The low of double negation:

$$\neg\neg A \Leftrightarrow A$$

Conjunction

- commutative
- associative
- idempotent

The set of formulae $\{A_1, A_2, \ldots, A_n\}$ where $A_1, A_2, \ldots, A_n \in Form$ is satisfiable if and only if $(A_1 \wedge A_2 \wedge \ldots \wedge A_n)$ formula satisfiable.
The set of formulae $\{A_1, A_2, \ldots, A_n\}$ where $A_1, A_2, \ldots, A_n \in Form$ is unsatisfiable if and only if $(A_1 \wedge A_2 \wedge \ldots \wedge A_n)$ unsatisfiable.
$\{A_1, A_2, \ldots, A_n\} \models A$ $(A_1, A_2, \ldots, A_n, A \in Form)$ if and only if $(A_1 \wedge A_2 \wedge \ldots \wedge A_n) \models A$.
$\{A_1, A_2, \ldots, A_n\} \models A$ $(A_1, A_2, \ldots, A_n, A \in Form)$ if and only if $((A_1 \wedge A_2 \wedge \ldots \wedge A_n) \wedge \neg A)$ is unsatisfiable.

Disjunction

The properties of the disjunction

- commutative
- associative
- idempotent
- $A \models (A \vee B)$
- $\{(A \vee B), \neg A\} \models B$

Connections of conjunction and disjunction

$$(A \wedge (B \vee A)) \Leftrightarrow A \qquad \text{and} \qquad (A \vee (B \wedge A)) \Leftrightarrow A$$

Distributivity rules

$$(A \vee (B \wedge C)) \Leftrightarrow ((A \vee B) \wedge (A \vee C))$$

$$(A \wedge (B \vee C)) \Leftrightarrow ((A \wedge B) \vee (A \wedge C))$$

De Morgan laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B) \qquad \text{and} \qquad \neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

Aristotelian principles of logic

the law of non-contradiction

A statement cannot be both true and false.

$$\models \neg(A \land \neg A)$$

the law of excluded middle

It is impossible to have a statement that is neither true nor false.

$$\models (A \lor \neg A)$$

Conjunction and disjunction chains

Let $A_1, A_2, \ldots, A_n \in Form$ where $n > 2$.

The conjunction chain with n members:

$$(A_1 \wedge A_2 \wedge \ldots \wedge A_n) \rightleftharpoons (A_1 \wedge (A_2 \wedge (A_3 \wedge (\ldots \wedge A_n) \ldots)))$$

The $(A_1 \wedge A_2 \wedge \ldots \wedge A_n)$ formula is true if and only if A_i is true for all $i = 1, 2, \ldots, n$.

The disjunction chain with n members:

$$(A_1 \vee A_2 \vee \ldots \vee A_n) \rightleftharpoons (A_1 \vee (A_2 \vee A_3 \vee (\ldots \vee A_n) \ldots)))$$

The $(A_1 \vee A_2 \vee \ldots \vee A_n)$ formula is false if and only if A_i is false for all $i = 1, 2, \ldots, n$.

Properties of the implication

$\models (A \supset A)$
Modus ponens: $\{(A \supset B), A\} \models B$
Modus tollens: $\{(A \supset B), \neg B\} \models \neg A$
Chain rule: $\{(A \supset B), (B \supset C)\} \models (A \supset C)$
Redukció ad abszurdum: $\{(A \supset B), (A \supset \neg B)\} \models \neg A$
Contraposition: $(A \supset B) \Leftrightarrow (\neg B \supset \neg A)$
$((A \wedge B) \supset C) \Leftrightarrow (A \supset (B \supset C))$
$\neg A \models (A \supset B)$
$B \models (A \supset B)$
$(A \supset \neg A) \models \neg A$
$(\neg A \supset A) \models A$
$(A \supset (B \supset C)) \Leftrightarrow ((A \supset B) \supset (A \supset C))$
$\models (A \supset (\neg A \supset B))$
$((A \vee B) \supset C) \Leftrightarrow ((A \supset C) \wedge (B \supset C))$
$\{A_1, A_2, \ldots, A_n\} \models A$ $(A_1, A_2, \ldots, A_n, A \in Form)$ if and only if $((A_1 \wedge A_2 \wedge \ldots \wedge A_n) \supset A)$ is valid.

Properties of the implication

The deduction theorem

Let $L^{(0)} = \langle LC, Con, Form \rangle$ be a *zero-order language*, and $A, B \in Form$ be tow arbitrary formula.

$$A \models B \text{ if and only if } \models (A \supset B).$$

Properties of the equivalence

Properties of the (material) equivalence:

$$\models (A \equiv A)$$

$$\models \neg(A \equiv \neg A)$$

Expressibility rules

Expressibility rules of the introduced connectives:

$$(A \supset B) \Leftrightarrow \neg(A \wedge \neg B)$$

$$(A \supset B) \Leftrightarrow (\neg A \vee B)$$

$$(A \wedge B) \Leftrightarrow \neg(A \supset \neg B)$$

$$(A \wedge B) \Leftrightarrow \neg(\neg A \vee \neg B)$$

$$(A \vee B) \Leftrightarrow (\neg A \supset B)$$

$$(A \vee B) \Leftrightarrow \neg(\neg A \wedge \neg B)$$

$$(A \equiv B) \Leftrightarrow ((A \supset B) \wedge (B \supset A))$$

Base

The *base* is such a set of connectives that any other connective can be expressed with those.

E.g.:

- $\{\neg, \supset\}$, where $(p \wedge q) \Leftrightarrow \neg(p \supset \neg q)$ and $(p \vee q) \Leftrightarrow (\neg p \supset q)$;
- $\{\neg, \wedge\}$;
- $\{\neg, \vee\}$.

Connectives that create singleton basis:

- Sheffer-Stroke (nand): $(p \mid q) \rightleftharpoons \neg(p \wedge q)$
- Pierce-Arrow (nor): $(p \downarrow q) \rightleftharpoons (\neg p \wedge \neg q)$